

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Metals and Non-metals

Formative-only topic in CBSE 2026-27: useful for school assessment; not year-end board assessed

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

Why do ionic compounds generally have high melting points?

Answer:

Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. A large amount of heat is needed to separate them, so melting points are usually high.

Question 2

What happens when an iron nail is placed in copper sulphate solution?

Answer:

Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. The nail gets a copper coating and the blue solution becomes pale green.

Question 3

State two methods of preventing rusting and explain the principle.

Answer:

Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. These methods interrupt the conditions needed for corrosion.

Question 4

What is the central idea or skill tested in Metals and Non-metals?

Answer:

The central focus is: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Metals and Non-metals.

Answer:

First point: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice.
Second point: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Metals and Non-metals lead to an important result or conclusion?

Answer:

Begin with the key idea: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Then connect it to the result: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Metals and Non-metals - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Metals and Non-metals?

Answer:

Use this supporting detail: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. A large amount of heat is needed to separate them, so melting points are usually high. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Metals and Non-metals.

Answer:

Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. The nail gets a copper coating and the blue solution becomes pale green. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Metals and Non-metals?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating.

Question 10

Compare the first and second important ideas in Metals and Non-metals.

Answer:

The first idea is: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. The second idea is: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Metals and Non-metals.

Answer:

Cause or starting point: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Effect or outcome: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Metals and Non-metals?

Answer:

A clear sequence is: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Next, Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Finally, Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating.

Metals and Non-metals - Continued

Question 13

Give a 40-50 word answer outline for an important question from Metals and Non-metals.

Answer:

Start with the direct answer: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Add one supporting point: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. End with the outcome or message: Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Metals and Non-metals?

Answer:

Use an accurate scientific statement: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Metals and Non-metals?

Answer:

Connect the idea to its reason or result. For instance: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Metals and Non-metals.

Answer:

A sound inference is that painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Metals and Non-metals.

Answer:

Apply the concept by first stating: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Then justify the response with: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Question 18

What would make an answer from Metals and Non-metals incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Also add: Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating.

Metals and Non-metals - Continued

Question 19

Prepare an Assertion-Reason style explanation based on Metals and Non-metals.

Answer:

Assertion: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice.

Reason: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Metals and Non-metals in two clear steps?

Answer:

Step 1: state the concept or event - Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Step 2: support it with the explanation or result - Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Question 21

What answer structure is suitable for a 3-mark question from Metals and Non-metals?

Answer:

Use three connected points: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Metals and Non-metals?

Answer:

Give an introduction, develop these key points - Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. - and close with: Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in Metals and Non-metals be applied to a new situation or case study?

Answer:

Identify the relevant principle first: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Then use the chapter's reasoning to explain the case: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Question 24

Which detail from Metals and Non-metals is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. Explain it in context instead of using it as an isolated statement.

Metals and Non-metals - Continued

Question 25

Write a brief concluding statement for an answer on Metals and Non-metals.

Answer:

A suitable conclusion is: Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for Metals and Non-metals.

Answer:

1. Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. 2. Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. 3. Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Metals and Non-metals before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. - and supported it with a reason, example, calculation, or outcome such as: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Question 28

Why does Metals and Non-metals remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating.

Question 29

Create a final 60-80 word answer plan for a long question from Metals and Non-metals.

Answer:

Opening: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice.

Development: Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Conclusion: Painting, oiling, or galvanising keeps oxygen and moisture away from iron; galvanising also gives a zinc coating. Use complete sentences and include all required evidence, steps, or references.

Question 30

What is one last-minute revision question you should ask yourself about Metals and Non-metals?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: Positive and negative ions are held by strong electrostatic attraction in a crystal lattice. and Iron is more reactive than copper and displaces it: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.